MGMT 490 Project2

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Hi, my name is Anthony Petrescu and for this project I decided to build a portfolio optimizer application and that was kind of path A for that, or for this project I should say. And so the goal of this tool was to allow investors or portfolio managers to kind of be able to input a custom set of stock tickers and be able to apply a set of real world constraints and then mathematically optimize assets that they chose based on their specific risk preferences that we have. And as you can see, I'm just running this locally using framework that's called Streamlet and it just turns my Python optimization code into an interactive web application.

So it's pretty easy to use. So let me walk through this app function. And the first thing I'll start with is just this left sidebar that we have here.

And it starts with our input assumptions. Currently I just have a mix of a variety of different tech giants here, Apple, Microsoft, Amazon, NVIDIA, so on and so forth. And this works with five to 15 different tickers and it's pulling historical data directly from Yahoo Finance.

So these are real, this is real data that is actually being pulled. So the next thing we can look at is the year of historical data. So I've set this application right now to pull five years of historical adjusted closing prices.

And once it pulls that data, it's going to end up calculating average annualized expected returns for each asset as well as the covariance matrix, meaning it'll calculate how their prices and movements correlate with each other. And that can be seen through this chart here that we have where we'll have the actual prices for each stock, which, you know, we can adjust to just see one of them at a time here, like Apple or all of them at a time. There we go.

But we can also see the normalized growth here by just changing this graph. So we'll see, you know, graph utilizing, again, those returns, the actual prices, and then, you know, that covariance matrix that we have as well. Switch back to this actual prices.

As we can see right now, the application we have is set to optimize the maximum Sharpe ratio. So you can see the resulting optimal weights below in which the app is going to tell us the, let's scroll, there we go, tell us the expected annual return, annual volatility, and the Sharpe ratio as well. And this is going to serve as our risk metric, and the Sharpe ratio is just going to be for the combined portfolio as a whole.

Now, if we end up scrolling down a little bit slow, we have this visualization of what's called an efficient frontier. And so this efficient frontier can be broken down into different pieces, right? We have these black diamonds, which are going to be our individual assets. You can see, you know, we have Nvidia, Microsoft, Amazon, and then the scattered dots represent 2000 completely different random combinations of, or random portfolio combinations, I should say.

And our red line traces the actual constrained efficient frontier in which, you know, our red star is going to indicate whether our current optimal, where our current optimal portfolio sits. All right. And so we'll talk a little bit of breaking that down and what the efficient frontier really is.

It's one of those questions I have a little bit later. We'll let that simmer for right now and actually move into our constraint settings. So a key feature kind of for any real world tool is to have different constraints, right? If, you know, let's say client is using this application, and they want to set, you know, a 20% max allocation for asset, then, you know, we want to see what ends up happening if we're able to adjust this down.

And that's what we're able to do with this application, right? We're able to adjust it down to, you know, our 20% metric, we're going to get, you know, a completely different optimal portfolio, as you can see with different expected annual return, different Sharpe ratio. And then as well, you know, a different efficient frontier is going to be calculated from our portfolio as well. So you can see here is once we calculated, we're essentially for forcibly diversifying our profile.

So our expected return is going to shift ever so slightly, and the optimal point on the efficient frontier, you know, is going to adjust and you know, that efficient frontier graph is going to look a little bit different to remain under the new constraint rounds that we've set at 20%. So, you know, what were those questions that I was talking about? Oh, I think there's a few important things to kind of delve into when looking at an application like this. And the first thing is, you know, what MAC optimization objective did I choose? And essentially, why did I choose to incorporate them, and I actually chose three different ones, I chose to use max Sharpe rate, or maximum Sharpe ratio, minimum volatility, and target return.

And I did that because I think they're all encompassing all three of them together. For a variety of different instances, I think maximum Sharpe ratio is one that tries to maximize return, but also minimize volatility. So it gets the best of both worlds with those other two.

And so Sharpe ratio is just going to be calculated using excess return generated per the unit of risk taken. And I ended up choosing this because for general investor, again, it fundamentally provides, say, best bang for your buck, I guess the best way to put it, right, it's going to create a portfolio that's going to theoretically maximize return, also penalizing excessive volatility, too. And that's generally the most robust starting point for applying any kind of client specific target returns, right, is just to be able to maximize return, while also, you know, reducing the amount of volatility we have, or at least penalizing it in this way.

But you know, what if we have an instance, you know, let's say for an elderly client who's entering retirement, and the important thing for them is just to minimize volatility, right, we want to ignore trying to just maximize returns and strictly just hope for the lowest possible risk possible, or the lowest risk possible, on the far left of the efficient frontier. And so we can change our optimization objective, and it's going to make, you know, our portfolio look a little bit different, right, you know, we're going to have a lower volatility, or sorry, we're going to have, you know, an expected annual return that, as you can see, is much lower, our Sharpe rate

is going to differ, but now we're minimizing volatility, and our efficient frontier is going to look a little bit different, where, you know, we were hunting for, you know, an optimum portfolio somewhere right here, we're now pushing further left to a much less risky portfolio. Another question I think is important to go over is how to estimate expected returns.

And so this application estimates expected returns using what's called the historical average annualized returns. And that's derived from a user selected timeframe, right. So it's that years of historical data, which we can change right now, we're doing five years of daily returns.

But it's also important to acknowledge that there are a little bit of limitations with this, right, past performance isn't always going to guarantee future results, it's really important concept to grasp in trading, and simply just averaging the last five years implicitly assumes that the market environment and any company's growth will remain identical moving forward. And then really, in any real institutional setting, it would be much better to use a more forward looking model like CAPM or, you know, fundamental DCF price targets to serve as expected return inputs, rather than just relying on a strictly rear view mirror for learning purposes and just in general, to be able to see historical data and see trends and where a company has been moving. I think that this kind of, you know, ticks off that or checks off that checkbox, I think makes it a really great tool for learning and just in general to see kind of variations of portfolio optimization.

Another I think important question is, you know, how are our constraints going to affect optimal portfolio, right? How do I know how to change different things? Why are these constraints going to change them? And really, constraints are just directly forcing this optimizing optimizer we have for our application to accept, we kind of grasp as a suboptimal portfolio in exchange for satisfying rule that we want to put in place, right. So for example, without any constraints that we have here, where we set it to, you know, 20% max weight per asset, we could have a maximum sharp optimizer, but it might dump, you know, 90% of, you know, investors' money into just NVIDIA, you know, if it had the best historical risk adjusted run. But, you know, by applying a maximum weight per asset constraint, which, you know, we have 20%, we're able to forcibly cap NVIDIA and the optimizer is now forced to distribute the remaining 80% into next best assets.

And that trade-off is, you know, pretty clear, overall portfolio sharp ratio is going to decrease, which is theoretically less optimal, but overall diversification is going to increase heavily, which is going to protect investors from catastrophic single stock downfalls, which is why, you know, diversification is so important. And the last piece that I was hinting at in kind of the last question is what effectively does this efficient frontier show, you know, what is this curve? And the efficient frontier curve just visualizes ultimate risk return trade-off. The X axis is going to be our risk, right? It's going to be volatility, as we can see here.

And the Y axis is going to be our reward. It's the best way to look at just the risk reward situation. It's going to, you know, our reward being expected return.

The red line is going to represent just in general, a boundary, which is the absolute maximum

possible return you can achieve for any given level of risk under our specific constraints that we put. An investor can use this to understand their choices, depending on their risk tolerance. You know, if an investor is highly risk averse, we're going to choose a portfolio resting, you know, on the far left side of the curve.

If they have a high risk tolerance, they will move further up and seek something that has a higher expected return, but maybe, you know, higher volatility. But, you know, the main concept to grasp with this, which I think is so important, is any portfolio that's going to plot below the line is fundamentally inefficient. It means that you're taking on risk without properly compensating, without being properly compensated with return for it.

And that's never really a good thing. And so that kind of sums up, you know, I guess, a walkthrough of this application, have a full driver framework documentation on, you know, the GitHub that is uploaded with this code, again, really easy to run using Streamlit. And then there is also an AI disclosure of just some of the things that I used with antigravity to help, you know, get visuals set up and just in general to be able to gather an application based off of some of my previous knowledge that I've used and learned in classes.

And so, you know, feel free to read through those and, you know, utilize this code for your own learning. And thank you for watching.

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